OBC-V2G22

PRODUCT REQUIREMENTS DOCUMENT (PRD)

800 V Bi-directional 22 kW On-Board Charger with Vehicle-to-Grid

Automotive EV Power Electronics | Rev A2 - Architecture, Atomic Requirements, Interfaces & Verification Baseline

Mission profile. The product converts grid AC power to an 800 V-class traction battery for charging and returns battery energy to the grid in approved V2G operating modes. The product baseline defines a 22 kW bidirectional architecture using a totem-pole PFC, isolated dual-active bridge, 1200 V SiC MOSFETs, AUTOSAR Classic Platform control, model-predictive ZVS optimization, ASIL-C safety intent, liquid cooling, CAN FD/SecOC, IP6K9K packaging, automated HiPot and cryptographic EOL provisioning.

Engineering baseline. This PRD defines the controlled product architecture, atomic requirements, interface ownership, safety mechanisms and objective verification evidence needed for design review and release. Values that depend on vehicle program, regional grid code, selected supplier data or installation are explicitly treated as controlled parameters until approved by the responsible engineering owner.

Definition	Controlled product interpretation
Product class	22 kW bidirectional on-board charger for an 800 V-class EV traction system with grid-interactive V2G capability.
Safety / assurance	Source baseline states ASIL-C compliance with hardware-based overcurrent lockouts. This PRD treats that statement as ASIL-C safety intent pending program-specific HARA/FSC/TSC allocation and objective ISO 26262 compliance evidence; ISO/SAE 21434 cybersecurity remains in scope.
Lifecycle mission	15-year vehicle lifecycle with NVM wear-leveling, thermal-cycle logging, EOL configuration control and evidence-based change impact.
Primary architecture	AC input protection/EMI filter -> bidirectional totem-pole PFC -> DC link -> isolated DAB -> HV battery; 1200 V SiC; liquid cold plate; AUTOSAR CP supervisory/control software.
Release philosophy	Hardware-independent product limits are separated from installation, supplier, calibration and vehicle-variant values. Every atomic requirement traces to objective verification evidence.

Release interpretation. Product requirements are separated from supplier-dependent, installation-dependent and calibration-controlled values. No planning target, typical datasheet value or unverified regional assumption shall be treated as a production limit without configuration-controlled evidence and approval.

PRODUCT GOVERNANCE, REQUIREMENT QUALITY & CHANGE CONTROL

Single-intent requirements, controlled parameters, explicit interface ownership and evidence-based release.

Safety mechanisms are allocated by physical response time and independence. Fast destructive faults such as semiconductor overcurrent, shoot-through or isolation-bias collapse are contained by autonomous hardware paths where application-software latency is not credible. Supervisory software owns classification, logging, degraded operation, service reporting and controlled recovery. Network commands shall never bypass local electrical, thermal or insulation limits.

Requirements in this PRD are intentionally atomic. A requirement is considered closed only when the released hardware revision, software build, calibration set, test equipment, environmental condition and vehicle/grid simulator configuration are identified in the evidence package. Planning targets and supplier typical values are not treated as production guarantees.

Parameter class	Examples / release rule
Immutable safety	Absolute current/voltage/temperature limits, hard gate inhibit, independent trip paths. Change requires safety impact analysis and affected DVP regression.
Variant controlled	AC region, connector type, battery range, cooling interface, power rating, software feature set. Released by signed variant code, BOM and calibration package.
Supplier dependent	SiC SOA/short-circuit withstand, isolated driver CMTI, capacitor life, magnetics insulation, sensor error. Use guaranteed production data and worst-case analysis.
Installation dependent	Grid impedance, PE/grounding, harness length, coolant conditions, packaging heat rejection and EMC environment. Captured in vehicle/charger ICD and integration DVP.
Calibration	Loop gains, thresholds, dead time, derate curves, timing, filters and mode maps. Authenticated, range checked and traceable to verification evidence.

Traceability rule. Each product requirement shall trace upward to a product mission, hazard/risk control or external interface obligation and downward to at least one verification artifact. Evidence reuse after a semiconductor, topology, software, protocol or installation change is permitted only after documented similarity analysis proves the changed parameter remains inside the assumptions of the linked analyses; otherwise the impacted test, model and safety argument are reopened.

SYSTEM BOUNDARY, ASSUMPTIONS & EXTERNAL OWNERSHIP

The OBC is one element of the vehicle charging function. It receives AC grid energy and vehicle/battery commands and exchanges high-voltage DC energy with the traction battery. Grid service disconnects, EVSE protection, battery contactors, coolant supply and vehicle supervisory permission remain external unless allocated by the vehicle ICD.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-SYS-001	The released OBC boundary shall identify ownership of AC protection, EVSE interlock, protective earth, HV battery contactors, coolant supply, wake/sleep authority and communication permissions.	System architecture and ICD review.
OBC-SYS-002	External mitigations credited in the safety case shall have explicit response-time, diagnostic and failure-independence assumptions.	Safety/ICD trace review.
OBC-SYS-003	Loss of an external permissive required for power transfer shall transition the OBC to the allocated non-energizing or controlled-energy state without increasing grid or battery current.	HIL/power-HIL permissive-loss test.
OBC-SYS-004	The source-defined 22 kW rating shall not be interpreted as a universal charge-time guarantee independent of battery usable energy, SOC window, grid limit and thermal conditions.	Product requirement review.

Verification, margin and future-change rule

Changes to EVSE architecture, battery contactor ownership, vehicle wake strategy or cooling boundary reopen the interface safety and sequencing analysis.

OPERATING MODES & STATE AUTHORITY

The product supports charge, V2G discharge, standby, sleep, precharge, diagnostic/service, controlled shutdown and fault-latched modes. Mode transitions are permission-based; power conversion is enabled only after sensing, communication validity, isolation/ground checks and local protection readiness meet the released criteria.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-MOD-010	Each released mode shall define entry conditions, exit conditions, command authority, power limits, timeout behavior and safe-state response.	State-machine review and HIL transition matrix.
OBC-MOD-011	A transition between charge and V2G shall pass through a controlled zero/limited-power state unless the released control analysis proves bumpless bidirectional reversal.	Power-HIL reversal test.
OBC-MOD-012	Reset, brownout or watchdog recovery shall not create an uncontrolled gate-enable or power-flow step.	Power-cycle and reset fault injection.
OBC-MOD-013	Conflicting vehicle, EVSE or grid-service commands shall resolve to a deterministic non-permissive authority rule.	Command-conflict HIL test.

Verification, margin and future-change rule

Any new operating mode, bidirectional transition policy or command owner requires state-machine regression and safety impact review.

AC INPUT, CONTACTOR INTERFACE & INRUSH CONTROL

The AC boundary includes connector/input pins, fusing/protection assumptions, line sensing, EMI filter, discharge paths and any internal relay/contactor functions. Normal, abnormal and transient input envelopes are variant-controlled because grid region and EVSE architecture differ.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-IN-020	Each AC variant shall define normal, abnormal and transient voltage/frequency envelopes used for component, insulation and control sizing.	Electrical ICD and bench sweep.
OBC-IN-021	Gate enable shall remain inhibited until AC sensing, DC-link state, control bias and protection health meet READY criteria.	Power sequencing test.
OBC-IN-022	Input inrush and DC-link energization shall remain within connector, fuse, relay and semiconductor stress limits for the released source impedance.	Inrush/precharge test with worst source impedance.
OBC-IN-023	Loss or bounce of an input relay/contactor signal shall not produce repeated destructive charging of the DC link.	Relay-bounce fault injection.

Verification, margin and future-change rule

AC-region, connector, EVSE or feeder-impedance changes reopen inrush, transient and EMC verification.

BIDIRECTIONAL TOTEM-POLE PFC POWER STAGE

The baseline architecture uses a totem-pole PFC with 1200 V SiC MOSFETs. Device selection is based on worst bus/transient voltage, switching overshoot, reverse conduction, short-circuit/fault energy and hot-junction margin rather than nominal grid voltage alone.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-PFC-030	PFC power devices shall retain released voltage, current, temperature and transient SOA at maximum AC/DC stress and commutation overshoot.	WCCA plus double-pulse/full-power switching test.
OBC-PFC-031	Commutation-loop inductance and gate profile shall meet the overshoot used in device derating analysis.	DPT and layout correlation.
OBC-PFC-032	Dead-time and reverse-conduction behavior shall prevent cross-conduction across the released voltage, temperature and tolerance envelope.	Worst-corner switching test.
OBC-PFC-033	A single switch/driver fault shall transition the stage to the allocated safe response before energy exceeds the protected component capability.	Fault-injection/short-circuit test.

Verification, margin and future-change rule

SiC device, package, gate driver or power-loop layout changes trigger switching, EMI, thermal and protection regression.

DC LINK, PRECHARGE, DISCHARGE & STORED ENERGY

The DC-link architecture stores enough energy to create hazardous stress after grid removal. Capacitance, ESR/ESL, bleed/discharge, precharge/inrush and voltage sensing therefore form a single controlled energy-management function. Release evidence shall include measured precharge current, DC-link rise time, discharge-to-service-safe timing, residual voltage after commanded shutdown and component-tolerance correlation. Abnormal interruption during precharge shall be tested to confirm that stored energy cannot create an uncontrolled re-energization sequence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-DCL-040	DC-link capacitance and ripple-current capability shall be sized for the released charge/V2G operating envelope and end-of-life loss budget.	WCCA, thermal and ripple-current validation.
OBC-DCL-041	The DC link shall not be declared safe-to-service until measured voltage is below the released threshold and any diagnostic uncertainty is accounted for.	Discharge timing and sensor-fault test.
OBC-DCL-042	Precharge/discharge elements shall remain within repetitive pulse-energy and temperature limits under restart and fault-recovery sequences.	Repeated-cycle energy test.
OBC-DCL-043	Invalid DC-link voltage data shall force a non-permissive state for gate enable and service discharge completion.	Frozen/stuck sensor injection.

Verification, margin and future-change rule

Capacitor technology, capacitance, precharge topology or service-discharge requirement changes reopen stored-energy and lifetime evidence.

ISOLATED DUAL-ACTIVE-BRIDGE DC/DC

The DAB provides galvanic isolation and bidirectional DC/DC conversion between the regulated DC link and the HV battery. Phase-shift modulation, transformer ratio, leakage inductance and device timing jointly determine power range, circulating current and ZVS behavior.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-DAB-050	The DAB shall meet the released bidirectional power envelope without exceeding semiconductor, transformer, inductor or capacitor current/voltage limits.	Full-power dyno/power-HIL sweep.
OBC-DAB-051	Phase-shift and dead-time commands shall be bounded to prevent loss of controllability or transformer flux imbalance.	SIL/HIL plus switching waveform validation.
OBC-DAB-052	Loss of ZVS shall be detected or bounded by a derated operating region such that device dissipation remains within thermal limits.	ZVS map correlation across corners.
OBC-DAB-053	A bridge-leg fault shall not transfer uncontrolled energy across the isolation barrier beyond the allocated protection response.	Fault containment test/analysis.

Verification, margin and future-change rule

Transformer ratio, leakage inductance, modulation strategy or device technology changes require power-map, ZVS, thermal and insulation regression.

ISOLATION TRANSFORMER, MAGNETICS & HIGH-FREQUENCY INSULATION

The transformer and filter magnetics are safety- and performance-critical because insulation stress, core loss, copper loss, parasitic capacitance and leakage inductance influence efficiency, common-mode current and DAB commutation. Manufacturing variation is included in the released model. The qualification basis shall separately account for DC working voltage, repetitive switching crest, common-mode excitation and manufacturing variation. Magnetics acceptance shall correlate design calculations with representative winding temperature, leakage inductance, insulation resistance and, where applicable, partial-discharge characterization.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-MAG-060	The isolation transformer shall meet released working/transient voltage, creepage/clearance, insulation-system and partial-discharge requirements across lifetime and environment.	HiPot/IR plus PD characterization where applicable.
OBC-MAG-061	Core flux density and copper/core hotspot temperature shall remain within released limits for maximum power, frequency and modulation corners.	Thermal mapping and magnetic WCCA.
OBC-MAG-062	Leakage and magnetizing inductance production variation shall remain within the control/ZVS assumptions.	EOL Lm/Llk measurement and correlation.
OBC-MAG-063	Parasitic capacitance/common-mode current shall remain within the EMC/leakage-current architecture.	Impedance extraction and conducted-emission/leakage test.

Verification, margin and future-change rule

Magnetic material, winding process, insulation system or supplier change triggers electrical, thermal, PD and EMC requalification.

ISOLATED GATE DRIVE, FAST OVERCURRENT & ACTIVE TURN-OFF

Destructive power-stage faults require autonomous gate-off independent of the AUTOSAR task. Gate drivers provide UVLO, CMTI margin, desaturation/overcurrent detection where appropriate, controlled turn-off and latched fault reporting. Protection timing shall be measured from physical fault onset through detection, propagation, gate discharge and current decay rather than only from a software timestamp. Worst-case tests shall include maximum DC-link voltage, hot junction, bias tolerance and commutation-loop inductance so the claimed device SOA margin is demonstrable.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-GD-070	Fault-onset-to-safe-current time shall remain within the selected power-device minimum guaranteed fault-withstand allocation.	Worst Vdc/Tj short-circuit test.
OBC-GD-071	Fast gate inhibit shall operate independently of the application software and network stack.	Processor-stall and bus-overload test.
OBC-GD-072	Active clamp or multi-stage turn-off shall keep device voltage and fault energy inside released SOA.	Fault waveform/energy integration.
OBC-GD-073	CMTI and off-state gate margin shall prevent false turn-on at worst qualified dv/dt.	DPT/CMTI susceptibility test.

Verification, margin and future-change rule

Gate-driver, isolated-bias, turn-off network, SiC package or layout changes trigger fast-fault and dv/dt regression.

VOLTAGE, CURRENT & TEMPERATURE METROLOGY

Accurate sensing supports grid current control, battery current control, power calculation, protection, diagnostics and thermal derating. Safety mechanisms do not rely on a single unqualified conversion chain when an independent fast path is required. The metrology budget shall allocate sensor gain/offset, isolation error, ADC reference, sampling skew, filter delay and temperature drift. Protection and control functions shall define how stale, saturated or implausible samples are identified so a plausible but corrupted value cannot silently increase charging or V2G authority.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-SNS-080	Each safety- or control-relevant sensor channel shall define range, accuracy, bandwidth, offset/gain drift, plausibility and diagnostic behavior over environment and lifetime.	Sensor WCCA and calibrated bench sweep.
OBC-SNS-081	Current/voltage measurement latency shall remain within control stability and protection timing allocations.	Timing correlation and step-response test.
OBC-SNS-082	Stale, frozen, saturated or implausible measurements shall propagate an invalid status rather than being silently reused.	Fault-injection test.
OBC-SNS-083	Thermal estimation shall remain conservative when one temperature sensor is unavailable or implausible.	Thermal sensor dropout test.

Verification, margin and future-change rule

Sensor topology, ADC reference, isolation amplifier or filtering changes require metrology, timing and safety revalidation.

CONTROL ARCHITECTURE, PFC LOOPS & DAB POWER CONTROL

The software baseline uses AUTOSAR Classic Platform with model-predictive control for ZVS optimization. Fast current loops, grid synchronization, DC-link control, DAB power control and thermal limits are partitioned so supervisory features cannot override local protection. Control release shall include plant-tolerance and source/load corner models correlated to hardware. Loop interaction between PFC current control, DC-link regulation and DAB power transfer shall be checked during startup, power reversal, current limiting and rapid battery-limit changes to avoid saturation-driven recovery transients.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-CTL-090	Control-loop sample/PWM timing shall be deterministic and synchronized to the released switching strategy.	Timing trace and SIL/HIL verification.
OBC-CTL-091	PFC current and DC-link loops shall meet released stability margins over grid, component and thermal tolerances.	Frequency-response and power-HIL sweep.
OBC-CTL-092	DAB power control shall bound phase shift, current and power commands across battery voltage and bidirectional flow.	Power-HIL corner matrix.
OBC-CTL-093	Control saturation and recovery shall use bounded rate/anti-windup behavior without creating a power step.	Saturation/recovery test.

Verification, margin and future-change rule

Switching frequency, control rate, loop ownership or plant parameter changes require stability and timing regression.

ZVS OPTIMIZATION & MODEL-PREDICTIVE CONTROL

Model-predictive ZVS optimization is treated as an efficiency/performance layer constrained by hard current, voltage and thermal limits. Invalid model inputs or prediction failure shall reduce capability rather than widen protection limits. Prediction confidence, parameter adaptation and fallback conditions shall be explicit calibration items. The fallback controller must remain stable and thermally acceptable outside the optimized ZVS region, and transitions into or out of predictive control shall be bumpless under voltage-ratio, load-step and temperature extremes.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-ZVS-100	The released ZVS optimization shall define valid operating region, model inputs, update rate and fallback behavior.	Model review and HIL boundary sweep.
OBC-ZVS-101	Prediction/model error shall not command phase shift or current outside independently enforced limits.	Model-mismatch fault injection.
OBC-ZVS-102	Loss of ZVS optimization shall transition to a bounded fallback modulation strategy with verified thermal capability.	Fallback efficiency/thermal test.
OBC-ZVS-103	Calibration updates affecting ZVS maps shall be authenticated, range checked and traceable to verification evidence.	Calibration-security and regression audit.

Verification, margin and future-change rule

Control-model, calibration or device-loss model changes trigger ZVS-map and thermal regression before release.

GRID SYNCHRONIZATION, POWER QUALITY & V2G EXPORT

Bidirectional operation requires valid grid synchronization and controlled real/reactive power behavior. Exact regional grid-code limits are installation/program parameters and remain TBC until the target market and certification plan are fixed.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-GRID-110	V2G power export shall be enabled only when grid voltage, frequency, phase validity and external permission meet the released regional criteria.	Grid-emulator conformance test.
OBC-GRID-111	Loss of grid synchronization or islanding permissive shall command a controlled non-exporting state within the allocated response.	Grid-disturbance/islanding test.
OBC-GRID-112	Charge and export current harmonics, power factor and DC injection shall meet the released market-specific limits.	Power-quality test with calibrated analyzer.
OBC-GRID-113	Reactive-power or grid-service commands shall remain subordinate to OBC thermal, current, battery and safety limits.	Command-priority power-HIL test.

Verification, margin and future-change rule

Grid-code, EVSE/DER profile or market change requires regional parameter release and power-quality regression.

BATTERY HV INTERFACE, CONTACTOR COORDINATION & REGEN ENERGY

The OBC does not unilaterally energize the battery bus. Battery contactor/precharge authority, pack voltage/current limits, insulation state and BMS permission are explicit interfaces. During V2G, battery discharge limits remain authoritative. Interface verification shall cover contactor-command loss, stale BMS limits, unexpected pack-voltage movement, insulation fault and rapid discharge-limit reduction during export. The OBC shall never infer permission from the absence of a negative message; enabling energy transfer requires explicit valid state and timeout-qualified authority.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-BAT-120	HV output enable shall require valid battery-voltage range, BMS permission, contactor state and insulation/safety readiness.	Vehicle-HIL/power-HIL sequence test.
OBC-BAT-121	OBC current command shall remain within the freshest valid BMS charge/discharge limits and local hardware limits.	Limit-step/freshness fault injection.
OBC-BAT-122	Loss of BMS communication shall transition to the released non-permissive current state without an uncontrolled DC-link excursion.	Communication-loss power test.
OBC-BAT-123	Contactor mismatch or unexpected HV presence shall block gate enable and raise a latched diagnostic according to the safety concept.	Contactor fault injection.

Verification, margin and future-change rule

Battery voltage class, BMS protocol, contactor ownership or pack precharge strategy changes reopen the HV interface and safety sequence.

FUNCTIONAL SAFETY, FTTI & INDEPENDENT PROTECTION

The source baseline states ASIL-C compliance with hardware-based overcurrent lockouts. For this engineering baseline, that statement is treated as ASIL-C safety intent until the program-specific HARA, FSC, TSC and associated objective compliance evidence are released. Safety mechanisms and timing ownership are therefore defined without inventing vehicle-level hazard classifications. Safety timing shall be allocated from physical fault onset to a bounded electrical state and shall include detector blanking, propagation, gate turn-off and stored-energy decay. External vehicle mitigations may be credited only when their interface timing, independence and failure behavior are explicitly controlled in the safety case.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-SAF-130	Each safety goal allocated to the OBC shall trace to one or more item requirements, mechanisms, diagnostics and verification evidence.	Safety traceability audit.
OBC-SAF-131	Fast overcurrent and shoot-through protection shall not depend on successful execution of the AUTOSAR application task.	Processor-stall plus fault test.
OBC-SAF-132	Safe/degraded states shall define whether AC, DC and gate paths are inhibited, discharged or limited after each detected high-severity fault.	Fault-response matrix test.
OBC-SAF-133	Fault detection plus reaction time shall remain inside the allocated FTTI/protected-component withstand budget at worst environment.	Timed fault-injection evidence.

Verification, margin and future-change rule

Any safety-goal, protection path, ASIL allocation or external mitigation change requires ISO 26262 impact analysis and affected DVP regression.

LIQUID COOLING, HOT-SOAK RESTART & DERATING

The product baseline requires a liquid-cooled cold plate and Tj below 150 C at full 22 kW load. That value is retained as a top-level ceiling; the released design shall use lower component-specific operating/derating limits where required by guaranteed lifetime data. Thermal validation shall map semiconductor junction estimates, magnetics temperatures, coolant inlet/outlet conditions and cold-plate interface resistance over the full duty cycle. Hot-soak restart and degraded-flow tests shall prove that derating acts early enough to preserve end-of-life thermal margin without relying on a single temperature sensor.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-THM-140	The cooling interface shall define coolant inlet temperature, flow/pressure boundary, cold-plate resistance and allowable degradation as controlled parameters.	Thermal ICD and bench characterization.
OBC-THM-141	Estimated semiconductor junction temperature shall remain below the source ceiling of 150 C and below any lower released device/lifetime limit at full 22 kW operation.	Thermal mapping and loss-model correlation.
OBC-THM-142	Hot-soak restart shall apply verified derating or inhibit when thermal margin is insufficient for requested power.	Hot-soak restart test.
OBC-THM-143	Loss or invalidity of coolant/temperature data shall reduce capability or shut down without increasing thermal stress.	Cooling-fault and sensor-fault test.

Verification, margin and future-change rule

Cold plate, TIM, coolant, power device or enclosure changes trigger thermal model correlation and degraded-cooling regression.

CYBERSECURITY, CAN FD / SecOC & COMMAND AUTHORITY

The product baseline specifies ISO/SAE 21434 compliance and CAN FD with SecOC. The security design protects software, calibration, diagnostics, V2G permissions and provisioning while maintaining independence of immediate local hardware protections. Security controls shall define command authority and freshness for charge/export permissions, diagnostic sessions and calibration changes. Loss of authentication infrastructure shall fail non-permissively for new power-transfer authority while autonomous overcurrent, overvoltage, isolation and thermal protections remain immediately available and independent of network services.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-CYB-150	Safety- or power-authority CAN FD messages shall define source, destination, freshness, integrity, timeout and non-permissive fallback.	SecOC/protocol negative testing.
OBC-CYB-151	Only authenticated approved software and calibration shall execute in a power-enabled state.	Modified-image/load negative test.
OBC-CYB-152	Loss or invalidity of cybersecurity services shall reject untrusted commands without delaying local hardware gate-off or thermal protection.	Concurrent cyber-fault/safety test.
OBC-CYB-153	Production key injection and certificate identity shall be traceable to the serialized hardware unit.	EOL provisioning audit.

Verification, margin and future-change rule

Cryptographic algorithm, key-management backend, CAN database or bootloader changes trigger TARA and security regression.

MECHANICAL PACKAGING, IP6K9K & HV CONNECTORS

The product baseline specifies an IP6K9K die-cast housing with vibration-resistant AC/DC headers. The enclosure must simultaneously support sealing, HV creepage/clearance, thermal conduction, pressure equalization, EMC bonding and serviceability. Packaging verification shall include tolerance stack-up at HV interfaces, sealing after thermal/vibration exposure, vent behavior, bonding continuity and creepage/clearance preservation after assembly variation. Connector keying and service access shall prevent incompatible HV/LV variants or maintenance actions from bypassing the intended isolation and interlock architecture.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-MEC-160	The released enclosure shall maintain IP6K9K sealing and connector retention after the defined vibration, thermal-cycle and pressure-wash sequence.	Sequential environmental/ingress test.
OBC-MEC-161	HV/AC connector keying and retention shall prevent incompatible mating and maintain required creepage/clearance after tolerance stack-up.	Mechanical/insulation inspection and test.
OBC-MEC-162	Enclosure bonding and gasket/fastener strategy shall support the EMC and protective-earth architecture throughout lifecycle wear.	Bonding resistance and EMC correlation.
OBC-MEC-163	Service disassembly/reassembly shall not silently invalidate sealing, thermal interface or HV safety without a controlled replacement process.	Serviceability/process validation.

Verification, margin and future-change rule

Housing, connector, vent, gasket, fastener or mounting changes require mechanical, ingress, EMC and thermal delta analysis.

MANUFACTURING, EOL HiPot & CRYPTOGRAPHIC PROVISIONING

The product baseline requires automated HiPot and end-of-line cryptographic key injection. Production release therefore couples electrical safety test, calibration, traceability, variant programming and security identity to the serialized unit. EOL acceptance shall use calibrated fixtures with controlled uncertainty and serialized genealogy linking hardware, software, calibration and cryptographic identity. A passing functional power test shall not mask process defects that consume insulation, thermal or current-measurement margin; critical characteristics require explicit process-capability and correlation evidence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-MFG-170	EOL test shall verify safety-critical power, sensing, gate-inhibit, communication and discharge functions with objective pass/fail limits.	EOL fixture qualification and golden-unit correlation.
OBC-MFG-171	Automated HiPot/insulation tests shall use configuration-controlled voltage, dwell, leakage limit and measurement uncertainty.	HiPot gauge capability and audit.
OBC-MFG-172	Cryptographic key/certificate injection shall be secure, serialized and protected against duplicate or unauthorized provisioning.	Provisioning process security audit.
OBC-MFG-173	Software trim shall not be used to mask an assembly defect that consumes safety or performance margin.	Process capability and calibration review.

Verification, margin and future-change rule

Fixture, test method, calibration algorithm, provisioning station or critical supplier process changes require manufacturing impact review and correlation.

15-YEAR LIFECYCLE, AGING COUNTERS & FIELD DATA

The product baseline specifies NVM wear-leveling and thermal-cycle logging for warranty tracking. Lifecycle evidence also covers power-device cycling, capacitor aging, magnetic insulation, sealing, connector wear, coolant interface degradation and software/configuration obsolescence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-LIFE-180	The released mission profile shall define energy throughput, thermal cycles, charge/V2G duty, environment and service events used for lifetime design.	Mission-profile review and reliability plan.
OBC-LIFE-181	NVM shall use wear management and retain critical event/cycle data over the released lifecycle with corruption recovery.	Endurance and power-interruption testing.
OBC-LIFE-182	Thermal-cycle and high-stress event counters shall be configuration-controlled and traceable to field diagnostics without being treated as sole proof of remaining life.	Field-data and diagnostic audit.
OBC-LIFE-183	Supplier or component obsolescence substitutions shall require similarity analysis against electrical, thermal, EMC, safety and lifetime assumptions.	Change-control review.

Verification, margin and future-change rule

Any mission-profile expansion or component substitution beyond qualified similarity bounds triggers targeted reliability and DVP regression.

PART II - DETAILED SUBSYSTEM PRDS & ENGINEERING ASSURANCE ANNEX

Expanded architecture, atomic requirements, fault containment, interfaces, DVP and lifecycle evidence.

This annex extends the product-level OBC baseline into subsystem requirements for design review, hardware/software allocation, ISO 26262 work products, cybersecurity analysis, interface definition, production release and lifecycle change control. Controlled parameters are used where the final vehicle, grid region, EVSE, battery, harness, coolant, software load or supplier configuration has not fixed a value.

Across Part II, every TBC shall have an owner and closure method. Evidence shall identify the released configuration, calibrated instrumentation and raw records needed to demonstrate the mechanism. Prior evidence may be reused only after documented similarity and impact review shows the affected stress and margin remain bounded; otherwise the impacted analysis and DVP shall be repeated.

Annex family	Primary closure intent
A01-A05	Power-on state, AC boundary, input transient, DC-link energy, PFC metrology
A06-A10	SiC switching, DAB energy transfer, magnetics, insulation, gate-drive protection
A11-A15	Timing, controls, ZVS, thermal, leakage/ground-fault
A16-A20	EMC, communications, V2G authority, diagnostics, cybersecurity
A21-A25	AUTOSAR, boot/update, calibration, manufacturing, EOL
A26-A30	Full-envelope DVP, environmental, lifetime, supplier change, release evidence

AC / VEHICLE POWER-ON STATE MACHINE

Detailed subsystem PRD / engineering assurance annex - control closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA01-01	The AC/Vehicle Power-On State Machine function shall configuration-control state authority, command and saturation limits, valid input conditions, timing bounds and recovery behavior for every released variant.	Architecture / ICD / configuration review.
OBCA01-02	Controller states, gate-enable conditions, precharge/discharge ownership, timeouts and invalid-input behavior shall be bounded across the released source/load and sensor-tolerance envelope.	SIL/HIL or power-HIL state transition sweep with timing, brownout and invalid-sensor injection.
OBCA01-03	Invalid state, timing, model or sensor information for AC/Vehicle Power-On State Machine shall reduce capability or inhibit gate enable; software shall not widen electrical, current, voltage, insulation or thermal protection limits.	State/timing boundary sweep with invalid-input, saturation and recovery injection.

Verification, margin and future-change rule

Incorrect sequencing can connect energy before line, DC-link, isolation, battery contactor or controller readiness is qualified. Closure evidence shall include: SIL/HIL or power-HIL state transition sweep with timing, brownout and invalid-sensor injection. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

AC SURGE, DIP, INTERRUPT & RESTART BOUNDARY

Detailed subsystem PRD / engineering assurance annex - electrical closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA02-01	The AC Surge, Dip, Interrupt and Restart Boundary design shall configuration-control its electrical operating envelope, component and interface limits, protection thresholds, tolerance assumptions and end-of-life margin for every released hardware variant.	Architecture / ICD / configuration review.
OBCA02-02	Normal, abnormal and transient AC conditions shall map to deterministic continue/derate/stop/restart behavior without exceeding stored-energy or device stress limits.	Programmable AC-source sweep with surge/dip/interruption and restart logging.
OBCA02-03	If qualified margin or required health data for AC Surge, Dip, Interrupt and Restart Boundary is unavailable, the OBC shall inhibit or derate the affected power path so the response cannot increase energy, current, voltage, insulation stress or thermal load.	Electrical boundary and fault-injection test with source/load transients recorded.

Verification, margin and future-change rule

Line disturbances can trigger repeated restart, uncontrolled DC-link energy or nuisance protection unless restart hysteresis and recovery are explicit. Closure evidence shall include: Programmable AC-source sweep with surge/dip/interruption and restart logging. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

INPUT EMI FILTER DAMPING & GRID IMPEDANCE STABILITY

Detailed subsystem PRD / engineering assurance annex - hardware closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA03-01	The Input EMI Filter Damping and Grid Impedance Stability design shall configuration-control its electrical operating envelope, component and interface limits, protection thresholds, tolerance assumptions and end-of-life margin for every released hardware variant.	Architecture / ICD / configuration review.
OBCA03-02	Input filter damping and control shall remain stable across the released grid/source impedance and component tolerance envelope.	Impedance analysis correlated to network analyzer and power-HIL weak-grid tests.
OBCA03-03	If qualified margin or required health data for Input EMI Filter Damping and Grid Impedance Stability is unavailable, the OBC shall inhibit or derate the affected power path so the response cannot increase energy, current, voltage, insulation stress or thermal load.	Representative hardware boundary test with waveform and protection-response evidence.

Verification, margin and future-change rule

The EMI filter and active PFC input impedance can interact with weak grids or EVSE impedance and create oscillation even when nominal conducted emissions pass. Closure evidence shall include: Impedance analysis correlated to network analyzer and power-HIL weak-grid tests. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

PFC COMMUTATION LOOP, DEAD TIME & REVERSE CONDUCTION

Detailed subsystem PRD / engineering assurance annex - power-stage closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA04-01	The PFC Commutation Loop, Dead Time and Reverse Conduction design shall configuration-control its electrical operating envelope, component and interface limits, protection thresholds, tolerance assumptions and end-of-life margin for every released hardware variant.	Architecture / ICD / configuration review.
OBCA04-02	Gate timing, loop inductance and reverse-conduction interval shall stay within device SOA and released loss/EMI limits over tolerance and temperature.	Double-pulse matrix with timing/temperature/layout correlation.
OBCA04-03	If qualified margin or required health data for PFC Commutation Loop, Dead Time and Reverse Conduction is unavailable, the OBC shall inhibit or derate the affected power path so the response cannot increase energy, current, voltage, insulation stress or thermal load.	Worst-case electrical switching / fault test with device-stress waveforms.

Verification, margin and future-change rule

Fast SiC switching can trade efficiency for overshoot, EMI and false turn-on. Dead time must remain safe over propagation mismatch and temperature. Closure evidence shall include: Double-pulse matrix with timing/temperature/layout correlation. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

PFC CURRENT METROLOGY & GRID CURRENT ACCURACY

Detailed subsystem PRD / engineering assurance annex - metrology closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA05-01	The PFC Current Metrology and Grid Current Accuracy measurement chain shall configuration-control sensed range, accuracy, latency, filtering, diagnostic coverage, reference dependencies and calibration limits for every released variant.	Architecture / ICD / configuration review.
OBCA05-02	Current measurement shall retain released accuracy, linearity and latency from minimum current through full bidirectional power and during commutation common-mode events.	Calibrated source/load sweep and dynamic step test.
OBCA05-03	If qualified margin or required health data for PFC Current Metrology and Grid Current Accuracy is unavailable, the OBC shall inhibit or derate the affected power path so the response cannot increase energy, current, voltage, insulation stress or thermal load.	Calibrated boundary sweep with offset, drift, saturation and latency injection.

Verification, margin and future-change rule

Current-sense error directly affects power quality, protection margin and exported V2G current. Closure evidence shall include: Calibrated source/load sweep and dynamic step test. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

DC-LINK CAPACITOR RIPPLE, ESR & END-OF-LIFE MARGIN

Detailed subsystem PRD / engineering assurance annex - lifetime closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA06-01	Lifecycle closure for DC-Link Capacitor Ripple, ESR and End-of-Life Margin shall configuration-control mission profile, aging/endurance assumptions, field counters, supplier/obsolescence dependencies, evidence-reuse limits and regression triggers.	Architecture / ICD / configuration review.
OBCA06-02	Capacitance, ESR and ripple-current margins shall include temperature, tolerance, mission profile and released end-of-life degradation.	WCCA, accelerated aging correlation and full-power ripple measurement.
OBCA06-03	If qualified margin or required health data for DC-Link Capacitor Ripple, ESR and End-of-Life Margin is unavailable, the OBC shall inhibit or derate the affected power path so the response cannot increase energy, current, voltage, insulation stress or thermal load.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

Capacitor ESR growth and capacitance loss can increase ripple, heating and control stress late in life. Closure evidence shall include: WCCA, accelerated aging correlation and full-power ripple measurement. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

DAB PHASE-SHIFT LIMITS & CIRCULATING CURRENT

Detailed subsystem PRD / engineering assurance annex - control closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA07-01	The DAB Phase-Shift Limits and Circulating Current function shall configuration-control state authority, command and saturation limits, valid input conditions, timing bounds and recovery behavior for every released variant.	Architecture / ICD / configuration review.
OBCA07-02	DAB modulation shall enforce bounded phase shift/current for every released battery/DC-link ratio and direction of power flow.	Power-HIL/full-power corner sweep with current waveform capture.
OBCA07-03	Invalid state, timing, model or sensor information for DAB Phase-Shift Limits and Circulating Current shall reduce capability or inhibit gate enable; software shall not widen electrical, current, voltage, insulation or thermal protection limits.	State/timing boundary sweep with invalid-input, saturation and recovery injection.

Verification, margin and future-change rule

A mathematically valid phase-shift command can still create excessive circulating current or transformer stress at light load or extreme voltage ratio. Closure evidence shall include: Power-HIL/full-power corner sweep with current waveform capture. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

TRANSFORMER FLUX BALANCE & SATURATION PROTECTION

Detailed subsystem PRD / engineering assurance annex - magnetics closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA08-01	The Transformer Flux Balance and Saturation Protection design shall configuration-control magnetic material, winding construction, flux/current limits, insulation interfaces, parasitics, thermal assumptions and manufacturing tolerances.	Architecture / ICD / configuration review.
OBCA08-02	The released design shall prevent or detect flux imbalance before core saturation or winding current exceeds protected limits.	Injected asymmetry test plus magnetic model correlation.
OBCA08-03	If qualified margin or required health data for Transformer Flux Balance and Saturation Protection is unavailable, the OBC shall inhibit or derate the affected power path so the response cannot increase energy, current, voltage, insulation stress or thermal load.	Magnetic/thermal boundary test correlated to representative hardware and fault injection.

Verification, margin and future-change rule

Control asymmetry, sensor offset or faulted gating can accumulate volt-seconds and saturate the transformer. Closure evidence shall include: Injected asymmetry test plus magnetic model correlation. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

HIGH-FREQUENCY INSULATION & PARTIAL-DISCHARGE MARGIN

Detailed subsystem PRD / engineering assurance annex - safety closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA09-01	The High-Frequency Insulation and Partial-Discharge Margin safety mechanism shall configuration-control hazard-relevant limits, independent detection and shutdown paths, FTTI assumptions, diagnostic coverage, safe/degraded states and external dependencies.	Architecture / ICD / configuration review.
OBCA09-02	Working voltage, repetitive transient crest, material system, creepage/clearance, PD margin and manufacturing variation shall be configuration-controlled.	HiPot/IR plus PDIV/PDEV or repetitive-impulse characterization where applicable.
OBCA09-03	Loss of required margin, diagnostic validity or protection readiness for High-Frequency Insulation and Partial-Discharge Margin shall force the allocated safe/degraded state through the independent protection path within the released response-time budget.	Independent protection fault-injection test with end-to-end response-time evidence.

Verification, margin and future-change rule

Repetitive SiC dv/dt can stress insulation differently from DC HiPot and may create partial-discharge aging without immediate functional symptoms. Closure evidence shall include: HiPot/IR plus PDIV/PDEV or repetitive-impulse characterization where applicable. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

ISOLATED BIAS SUPPLY UVLO & COMMON-CAUSE FAULTS

Detailed subsystem PRD / engineering assurance annex - safety closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA10-01	The Isolated Bias Supply UVLO and Common-Cause Faults safety mechanism shall configuration-control hazard-relevant limits, independent detection and shutdown paths, FTTI assumptions, diagnostic coverage, safe/degraded states and external dependencies.	Architecture / ICD / configuration review.
OBCA10-02	Isolated bias architecture shall default gates OFF on undervoltage and contain a single bias fault within the allocated protection independence.	Bias brownout, short/open and common-cause fault injection.
OBCA10-03	Loss of required margin, diagnostic validity or protection readiness for Isolated Bias Supply UVLO and Common-Cause Faults shall force the allocated safe/degraded state through the independent protection path within the released response-time budget.	Independent protection fault-injection test with end-to-end response-time evidence.

Verification, margin and future-change rule

A gate-driver bias fault can force a device into linear operation or defeat both directions of a bridge if common supplies are not contained. Closure evidence shall include: Bias brownout, short/open and common-cause fault injection. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

AUTOSAR ISR TIMING, PWM SYNCHRONIZATION & WCET

Detailed subsystem PRD / engineering assurance annex - software closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA11-01	The AUTOSAR ISR Timing, PWM Synchronization and WCET software function shall configuration-control task ownership, scheduling and WCET bounds, data validity, state transitions, diagnostics, watchdog interaction and recovery behavior.	Architecture / ICD / configuration review.
OBCA11-02	Measured WCET, interrupt latency and PWM/ADC synchronization shall retain release-controlled timing reserve under maximum communication, diagnostic and fault-interrupt load.	Instrumented WCET stress test and HIL timing injection.
OBCA11-03	Invalid state, timing, model or sensor information for AUTOSAR ISR Timing, PWM Synchronization and WCET shall reduce capability or inhibit gate enable; software shall not widen electrical, current, voltage, insulation or thermal protection limits.	SIL/HIL timing and invalid-data injection with WCET / task-overload evidence.

Verification, margin and future-change rule

Timing jitter or overload can destabilize current loops or delay safety classification even when average CPU utilization is acceptable. Closure evidence shall include: Instrumented WCET stress test and HIL timing injection. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

PFC LOOP STABILITY ACROSS WEAK GRID & DC-LINK CORNERS

Detailed subsystem PRD / engineering assurance annex - control closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA12-01	The PFC Loop Stability Across Weak Grid and DC-Link Corners function shall configuration-control state authority, command and saturation limits, valid input conditions, timing bounds and recovery behavior for every released variant.	Architecture / ICD / configuration review.
OBCA12-02	Loop gain/phase margins shall remain within released stability criteria for all approved grid/source and power-flow corners.	Frequency-response identification and power-HIL sweep.
OBCA12-03	Invalid state, timing, model or sensor information for PFC Loop Stability Across Weak Grid and DC-Link Corners shall reduce capability or inhibit gate enable; software shall not widen electrical, current, voltage, insulation or thermal protection limits.	State/timing boundary sweep with invalid-input, saturation and recovery injection.

Verification, margin and future-change rule

Outer DC-link and inner current loops must remain stable across source impedance, line frequency, DC-link loading and parameter tolerance. Closure evidence shall include: Frequency-response identification and power-HIL sweep. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

MODEL-PREDICTIVE ZVS ESTIMATION ERROR & FALLBACK

Detailed subsystem PRD / engineering assurance annex - control closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA13-01	The Model-Predictive ZVS Estimation Error and Fallback function shall configuration-control state authority, command and saturation limits, valid input conditions, timing bounds and recovery behavior for every released variant.	Architecture / ICD / configuration review.
OBCA13-02	The MPC layer shall be bounded by independent electrical/thermal limits and enter a validated fallback when model confidence or required inputs are invalid.	Model mismatch, stale-input and fallback thermal tests.
OBCA13-03	Invalid state, timing, model or sensor information for Model-Predictive ZVS Estimation Error and Fallback shall reduce capability or inhibit gate enable; software shall not widen electrical, current, voltage, insulation or thermal protection limits.	State/timing boundary sweep with invalid-input, saturation and recovery injection.

Verification, margin and future-change rule

Model error can move the commanded operating point out of ZVS or into higher current/loss regions. Closure evidence shall include: Model mismatch, stale-input and fallback thermal tests. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

COLD-PLATE THERMAL RESISTANCE & HOT-SPOT CORRELATION

Detailed subsystem PRD / engineering assurance annex - thermal closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA14-01	The Cold-Plate Thermal Resistance and Hot-Spot Correlation thermal design shall configuration-control heat-source limits, sensor locations, interface resistance, cooling/environment boundary, derating thresholds and end-of-life thermal margin.	Architecture / ICD / configuration review.
OBCA14-02	Junction, magnetics and capacitor hot spots shall remain below released end-of-life limits for the full duty cycle including degraded cooling/interface resistance.	Thermal mapping and faulted-cooling tests correlated to an end-of-life model.
OBCA14-03	If required thermal margin or temperature validity for Cold-Plate Thermal Resistance and Hot-Spot Correlation is unavailable, the OBC shall derate or inhibit power transfer before semiconductor, magnetics, insulation or coolant-interface limits are exceeded.	Thermal boundary and degraded-cooling test with derating response recorded.

Verification, margin and future-change rule

Beginning-of-life thermal margin can disappear with TIM pump-out, coolant degradation or assembly variation. Closure evidence shall include: Thermal mapping and faulted-cooling tests correlated to an end-of-life model. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

LEAKAGE CURRENT, PE / GROUND-FAULT & RESIDUAL CURRENT

Detailed subsystem PRD / engineering assurance annex - safety closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA15-01	The Leakage Current, PE/Ground-Fault and Residual Current safety mechanism shall configuration-control hazard-relevant limits, independent detection and shutdown paths, FTTI assumptions, diagnostic coverage, safe/degraded states and external dependencies.	Architecture / ICD / configuration review.
OBCA15-02	The released design shall define PE/bonding, common-mode capacitance, leakage/residual-current behavior and response to insulation/ground faults for the vehicle/EVSE architecture.	Leakage/residual-current and insulation-fault injection testing.
OBCA15-03	Loss of required margin, diagnostic validity or protection readiness for Leakage Current, PE/Ground-Fault and Residual Current shall force the allocated safe/degraded state through the independent protection path within the released response-time budget.	Independent protection fault-injection test with end-to-end response-time evidence.

Verification, margin and future-change rule

High-frequency common-mode current can affect touch/leakage behavior and external residual-current detection. Closure evidence shall include: Leakage/residual-current and insulation-fault injection testing. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

CONDUCTED / RADIATED EMC & CABLE COMMON-MODE CURRENT

Detailed subsystem PRD / engineering assurance annex - EMC closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA16-01	The Conducted/Radiated EMC and Cable Common-Mode Current EMC design shall configuration-control switching/traffic conditions, cable and return-path assumptions, filter/bonding implementation, susceptibility limits and monitored functional behavior.	Architecture / ICD / configuration review.
OBCA16-02	EMI filters, shielding, bonding and edge-rate control shall meet released emissions/susceptibility limits without consuming protection or thermal margin.	CISPR/vehicle-level EMC test plus current-probe correlation.
OBCA16-03	If qualified margin or required health data for Conducted/Radiated EMC and Cable Common-Mode Current is unavailable, the OBC shall inhibit or derate the affected power path so the response cannot increase energy, current, voltage, insulation stress or thermal load.	EMC / transient immunity test with power-quality, protection and communication monitoring.

Verification, margin and future-change rule

SiC edge rates couple through AC, HV, coolant/structure and communication paths, so local switch-node optimization alone is insufficient. Closure evidence shall include: CISPR/vehicle-level EMC test plus current-probe correlation. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

CAN FD SIGNAL OWNERSHIP, FRESHNESS & TIMEOUT

Detailed subsystem PRD / engineering assurance annex - interface closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA17-01	The CAN FD Signal Ownership, Freshness and Timeout interface shall configuration-control signal ownership, units/range, update rate, freshness/integrity checks, timeout behavior, authority and non-permissive fallback.	Architecture / ICD / configuration review.
OBCA17-02	Every safety- or performance-relevant CAN FD signal shall define source, units/range, update rate, SecOC/freshness rule, timeout and non-permissive fallback.	Protocol conformance with stale/range/integrity fault injection.
OBCA17-03	Invalid, stale or conflicting information for CAN FD Signal Ownership, Freshness and Timeout shall be treated as non-permissive; charge/export capability shall be limited or stopped without increasing grid current, battery current, DC-link stress or thermal load.	Protocol / interface fault injection covering stale, range, integrity, timeout and authority behavior.

Verification, margin and future-change rule

Stale or ambiguously owned battery/grid permission can be more dangerous than complete communication loss. Closure evidence shall include: Protocol conformance with stale/range/integrity fault injection. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

V2G AUTHORIZATION & CHARGE / EXPORT REVERSAL

Detailed subsystem PRD / engineering assurance annex - system closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA18-01	The V2G Authorization and Charge/Export Reversal system function shall configuration-control external authority, operating-state prerequisites, direction/mode transition rules, interlocks, fault response and recovery criteria.	Architecture / ICD / configuration review.
OBCA18-02	Charge-to-export and export-to-charge transitions shall use a configuration-controlled authorization sequence and bounded power slew.	Grid/battery power-HIL reversal matrix.
OBCA18-03	Invalid, stale or conflicting information for V2G Authorization and Charge/Export Reversal shall be treated as non-permissive; charge/export capability shall be limited or stopped without increasing grid current, battery current, DC-link stress or thermal load.	End-to-end mode-transition and authorization fault injection with energy-flow response recorded.

Verification, margin and future-change rule

A reversal in energy direction must coordinate grid authorization, BMS discharge permission, DC-link control and thermal capability. Closure evidence shall include: Grid/battery power-HIL reversal matrix. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

FAULT CLASSIFICATION, LATCHING & RE-ARM POLICY

Detailed subsystem PRD / engineering assurance annex - diagnostic closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA19-01	The Fault Classification, Latching and Re-Arm Policy design shall configuration-control the operating envelope, interfaces, ownership, limits, assumptions, fault response and objective release evidence for every released variant.	Architecture / ICD / configuration review.
OBCA19-02	Each diagnostic shall define severity, debounce, latch behavior, power-limit effect, service data and explicit re-arm authority.	Fault-injection and repeated-recovery test.
OBCA19-03	If qualified margin or required health data for Fault Classification, Latching and Re-Arm Policy is unavailable, the OBC shall inhibit or derate the affected power path so the response cannot increase energy, current, voltage, insulation stress or thermal load.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

Automatic re-arm can convert an intermittent short, insulation fault or thermal problem into repeated destructive cycling. Closure evidence shall include: Fault-injection and repeated-recovery test. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

SECURE BOOT, SIGNED UPDATE & CALIBRATION PROTECTION

Detailed subsystem PRD / engineering assurance annex - cyber closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA20-01	The Secure Boot, Signed Update and Calibration Protection design shall configuration-control the operating envelope, interfaces, ownership, limits, assumptions, fault response and objective release evidence for every released variant.	Architecture / ICD / configuration review.
OBCA20-02	Only authenticated approved software/calibration shall execute in a power-enabled state; recovery shall preserve local hardware protections.	Modified-image, rollback and corrupted-calibration negative tests.
OBCA20-03	If qualified margin or required health data for Secure Boot, Signed Update and Calibration Protection is unavailable, the OBC shall inhibit or derate the affected power path so the response cannot increase energy, current, voltage, insulation stress or thermal load.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

Compromised software or calibration could change power limits, dead time, protection thresholds or V2G authority. Closure evidence shall include: Modified-image, rollback and corrupted-calibration negative tests. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

AUTOSAR PARTITIONING, WATCHDOG & FREEDOM FROM INTERFERENCE

Detailed subsystem PRD / engineering assurance annex - software closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA21-01	The AUTOSAR Partitioning, Watchdog and Freedom From Interference software function shall configuration-control task ownership, scheduling and WCET bounds, data validity, state transitions, diagnostics, watchdog interaction and recovery behavior.	Architecture / ICD / configuration review.
OBCA21-02	Task/resource allocation, watchdog servicing and memory/CPU protection shall preserve control timing and fault response under worst defined load.	WCET, overload and watchdog fault tests.
OBCA21-03	Invalid state, timing, model or sensor information for AUTOSAR Partitioning, Watchdog and Freedom From Interference shall reduce capability or inhibit gate enable; software shall not widen electrical, current, voltage, insulation or thermal protection limits.	SIL/HIL timing and invalid-data injection with WCET / task-overload evidence.

Verification, margin and future-change rule

Diagnostics, communication or logging bursts must not starve fast control or safety supervision. Closure evidence shall include: WCET, overload and watchdog fault tests. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

NVM WEAR-LEVELING, EVENT LOGGING & POWER-FAIL INTEGRITY

Detailed subsystem PRD / engineering assurance annex - lifecycle closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA22-01	Lifecycle closure for NVM Wear-Leveling, Event Logging and Power-Fail Integrity shall configuration-control mission profile, aging/endurance assumptions, field counters, supplier/obsolescence dependencies, evidence-reuse limits and regression triggers.	Architecture / ICD / configuration review.
OBCA22-02	NVM architecture shall bound write endurance, preserve critical first-fault data and recover consistently from interrupted writes.	Accelerated endurance and power-cut test.
OBCA22-03	If lifecycle, reliability, similarity or change-impact evidence for NVM Wear-Leveling, Event Logging and Power-Fail Integrity is incomplete, the affected variant or second source shall not be approved until the required analysis and targeted regression are closed.	Lifecycle / reliability / similarity review with approved change-impact analysis and targeted regression evidence.

Verification, margin and future-change rule

Frequent fault, thermal-cycle and energy counters can wear storage or corrupt first-fault evidence during abrupt power loss. Closure evidence shall include: Accelerated endurance and power-cut test. A change to the process, qualification basis, supplier or acceptance method shall reopen the affected release gate before equivalence is claimed.

CALIBRATION LIMITS, VARIANT CODING & CHANGE TRACEABILITY

Detailed subsystem PRD / engineering assurance annex - configuration closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA23-01	The Calibration Limits, Variant Coding and Change Traceability design shall configuration-control the operating envelope, interfaces, ownership, limits, assumptions, fault response and objective release evidence for every released variant.	Architecture / ICD / configuration review.
OBCA23-02	Power rating, grid region, battery range, thermal limits and control maps shall be signed/range-checked and matched to the serialized BOM/hardware revision.	Variant mismatch and calibration-authentication test.
OBCA23-03	If qualified margin or required health data for Calibration Limits, Variant Coding and Change Traceability is unavailable, the OBC shall inhibit or derate the affected power path so the response cannot increase energy, current, voltage, insulation stress or thermal load.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

A wrong calibration or variant code can create a fully functional but unsafe configuration. Closure evidence shall include: Variant mismatch and calibration-authentication test. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

MANUFACTURING PROCESS CAPABILITY & CRITICAL CHARACTERISTICS

Detailed subsystem PRD / engineering assurance annex - manufacturing closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA24-01	Production release for Manufacturing Process Capability and Critical Characteristics shall configuration-control critical characteristics, process capability, fixture/test coverage, measurement uncertainty, genealogy, calibration/provisioning and acceptance limits.	Architecture / ICD / configuration review.
OBCA24-02	Critical characteristics, process capability, measurement uncertainty and trim boundaries shall be configuration-controlled; trim shall not conceal structural defects.	Gauge capability, golden-unit and process-correlation evidence.
OBCA24-03	If process capability, measurement traceability or required production evidence for Manufacturing Process Capability and Critical Characteristics is incomplete, production release of the affected variant shall remain blocked until the defined acceptance evidence is closed and approved.	Process-capability, gauge/fixture correlation, genealogy and approved production-release evidence.

Verification, margin and future-change rule

Process drift in creepage, TIM thickness, torque, solder quality, shunt value or magnetics can consume design margin while EOL still appears nominal. Closure evidence shall include: Gauge capability, golden-unit and process-correlation evidence. A change to the process, qualification basis, supplier or acceptance method shall reopen the affected release gate before equivalence is claimed.

EOL HiPot, FUNCTIONAL POWER TEST & KEY INJECTION

Detailed subsystem PRD / engineering assurance annex - manufacturing closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA25-01	Production release for EOL HiPot, Functional Power Test and Key Injection shall configuration-control critical characteristics, process capability, fixture/test coverage, measurement uncertainty, genealogy, calibration/provisioning and acceptance limits.	Architecture / ICD / configuration review.
OBCA25-02	The EOL flow shall define HiPot/IR, low/full-power functional coverage as applicable, gate inhibit, sensing, discharge, network and secure provisioning acceptance criteria.	Qualified EOL fixture with GR&R/golden-unit correlation.
OBCA25-03	If process capability, measurement traceability or required production evidence for EOL HiPot, Functional Power Test and Key Injection is incomplete, production release of the affected variant shall remain blocked until the defined acceptance evidence is closed and approved.	Process-capability, gauge/fixture correlation, genealogy and approved production-release evidence.

Verification, margin and future-change rule

EOL must prove insulation, core power functions, protection and unique security identity without overstressing the new unit. Closure evidence shall include: Qualified EOL fixture with GR&R/golden-unit correlation. A change to the process, qualification basis, supplier or acceptance method shall reopen the affected release gate before equivalence is claimed.

FULL-ENVELOPE CHARGE POWER DVP

Detailed subsystem PRD / engineering assurance annex - verification closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA26-01	Verification release for Full-Envelope Charge Power DVP shall configuration-control representative boundary conditions, calibrated instrumentation, raw-data retention, configuration identity and objective pass/fail criteria.	Architecture / ICD / configuration review.
OBCA26-02	The DVP shall cover released charge corners, thermal steady-state, transients, credible faults and objective pass/fail criteria fixed before execution.	Configuration-controlled AC/grid emulator and battery emulator test matrix.
OBCA26-03	If the test matrix, instrumentation traceability, raw evidence or objective pass/fail disposition for Full-Envelope Charge Power DVP is incomplete, the verification gate shall remain open and no production-equivalence claim shall be made.	Configuration-controlled DVP report with calibrated instrumentation, raw records and explicit pass/fail disposition.

Verification, margin and future-change rule

Nominal-room tests can miss combined line, battery, coolant and tolerance conditions that set real charging limits. Closure evidence shall include: Configuration-controlled AC/grid emulator and battery emulator test matrix. A change to the process, qualification basis, supplier or acceptance method shall reopen the affected release gate before equivalence is claimed.

FULL-ENVELOPE V2G / REVERSE-POWER DVP

Detailed subsystem PRD / engineering assurance annex - verification closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA27-01	Verification release for Full-Envelope V2G/Reverse-Power DVP shall configuration-control representative boundary conditions, calibrated instrumentation, raw-data retention, configuration identity and objective pass/fail criteria.	Architecture / ICD / configuration review.
OBCA27-02	The DVP shall cover all released V2G regions, reversals, grid disturbances, communication faults and degraded thermal states.	Bidirectional grid/battery emulator DVP with raw waveform retention.
OBCA27-03	If the test matrix, instrumentation traceability, raw evidence or objective pass/fail disposition for Full-Envelope V2G/Reverse-Power DVP is incomplete, the verification gate shall remain open and no production-equivalence claim shall be made.	Configuration-controlled DVP report with calibrated instrumentation, raw records and explicit pass/fail disposition.

Verification, margin and future-change rule

Reverse power stresses control authority, grid synchronization, battery discharge limits and some passive elements differently from charging. Closure evidence shall include: Bidirectional grid/battery emulator DVP with raw waveform retention. A change to the process, qualification basis, supplier or acceptance method shall reopen the affected release gate before equivalence is claimed.

ENVIRONMENT, VIBRATION, INGRESS & SEQUENTIAL STRESS

Detailed subsystem PRD / engineering assurance annex - qualification closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA28-01	Environmental qualification shall configuration-control temperature, vibration, ingress, installation interfaces, sequential-stress order, post-stress electrical acceptance limits and released variant identity.	Environmental qualification plan and installation-interface review.
OBCA28-02	The qualification plan shall define temperature, vibration, ingress and other vehicle-environment categories from the actual installation and verify post-stress electrical margins.	Sequential environmental stress plus post-stress electrical retest.
OBCA28-03	If the environmental qualification matrix, sequential-stress evidence or post-stress electrical margin is incomplete for a hardware or installation variant, that variant shall not be released until the required qualification evidence is closed and approved.	Environmental qualification closure review with sequential-stress records and post-stress electrical retest.

Verification, margin and future-change rule

Sealing, insulation, solder, connectors and thermal interfaces can degrade only after combined environment sequences. Closure evidence shall include: Sequential environmental + electrical retest. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

MISSION PROFILE, POWER CYCLING & 15-YEAR RELIABILITY

Detailed subsystem PRD / engineering assurance annex - lifecycle closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA29-01	Lifecycle closure for Mission Profile, Power Cycling and 15-Year Reliability shall configuration-control mission profile, aging/endurance assumptions, field counters, supplier/obsolescence dependencies, evidence-reuse limits and regression triggers.	Architecture / ICD / configuration review.
OBCA29-02	The released mission profile shall drive power-cycling, capacitor, magnetics, connector/seal and NVM life models with documented confidence and margin.	Reliability prediction plus accelerated/mission-profile correlation.
OBCA29-03	If lifecycle, reliability, similarity or change-impact evidence for Mission Profile, Power Cycling and 15-Year Reliability is incomplete, the affected variant or second source shall not be approved until the required analysis and targeted regression are closed.	Lifecycle / reliability / similarity review with approved change-impact analysis and targeted regression evidence.

Verification, margin and future-change rule

A single full-power efficiency point does not prove 15-year life; cycling amplitude and dwell dominate many failure mechanisms. Closure evidence shall include: Reliability prediction plus accelerated/mission-profile correlation. A change to the process, qualification basis, supplier or acceptance method shall reopen the affected release gate before equivalence is claimed.

SUPPLIER CHANGE, SECOND SOURCE & RELEASE EVIDENCE CLOSURE

Detailed subsystem PRD / engineering assurance annex - release closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA30-01	Supplier and second-source release shall configuration-control component identity, guaranteed limits, approved similarity assumptions, affected analyses and tests, evidence-reuse boundaries and production-equivalence status for every released variant.	Supplier change-control, configuration and production-equivalence review.
OBCA30-02	Evidence reuse after supplier/topology/software changes shall be allowed only inside documented similarity bounds; affected margins and tests shall be reopened otherwise.	Change-impact review and targeted regression DVP.
OBCA30-03	A supplier or second-source component shall not be approved for production equivalence until documented similarity analysis, affected margin review and required regression DVP are closed.	Supplier change-impact review, similarity analysis and targeted regression DVP.

Verification, margin and future-change rule

Second-source or technology substitutions can preserve nominal datasheet ratings while changing switching, thermal, insulation or EMC behavior. Closure evidence shall include: Change-impact review and targeted regression DVP. Any design or interface change that alters these verified assumptions shall trigger targeted regression before unrestricted release.

REV A2 RELEASE CLOSURE - ENGINEERING ASSURANCE BASELINE

Requirements, evidence, configuration and future-change closure.

The OBC-V2G22 Rev A2 baseline combines the product-level architecture with a detailed system PRD and subsystem engineering-assurance annex. Release requires closure of variant-specific TBC values, reconciliation with the vehicle safety/cybersecurity lifecycle and objective evidence for each atomic requirement. No requirement is considered complete only because a related higher-level paragraph exists; completion is tied to the specified test, analysis, inspection, timing trace or configuration evidence.

Gate	Evidence required / release blocker
Architecture / ICD	Released limits, variants, state/authority model and interfaces. Blocker: unbounded or conflicting ownership.
Safety / protection	Closed fault paths, timing/SOA and ISO 26262 work-product traceability. Blocker: open high-severity mechanism or unverified FTTI.
HW / SW	WCCA, control stability, timing/WCET, calibration bounds and configuration integrity. Blocker: unverified derived requirement or margin deficit.
Verification	DVP with calibrated instrumentation, raw evidence and explicit pass/fail. Blocker: failed/waived critical test without approved rationale.
Manufacturing	Critical characteristics, HiPot/EOL capability, programming/provisioning traceability. Blocker: process cannot demonstrate repeatable margin.
Lifecycle	Mission profile, aging model, field counters, service limits and obsolescence controls. Blocker: unsupported 15-year life or change equivalence.

Future-proof rule. Reuse of existing evidence after a design, supplier, vehicle-interface or standards change is allowed only inside documented similarity bounds. Any change affecting electrical stress, control stability, timing, insulation, cybersecurity authority, thermal resistance, reliability or interface ownership shall trigger targeted impact analysis and regression DVP before production equivalence is claimed.

Candidate standards / work products to confirm by program

- Baseline-defined: ISO/SAE 21434; CAN FD with SecOC; ASIL-C safety intent and hardware overcurrent lockouts.
- Automotive safety/process candidates: ISO 26262 item/HARA/FSC/TSC, ASPICE work products and program-specific OEM requirements.
- Charging/grid candidates for confirmation by market: ISO 15118 / IEC 61851 family and applicable regional V2G/grid-code requirements.
- Vehicle electrical/environment/EMC candidates: applicable ISO 16750 / CISPR 25 / UN ECE R10 / OEM HV and insulation specifications.
- These candidate references require program confirmation; edition, scope and applicability must be confirmed by the responsible compliance owner.

RELEASE INTERPRETATION BOUNDARY

Any numeric value, regional grid-code constraint, supplier characteristic, vehicle/BMS interface limit or manufacturing threshold not explicitly released in this PRD remains a controlled program parameter. It shall be closed through the applicable ICD, parameter register, safety analysis, WCCA, DVP or production release evidence before it is used as a compliance claim.